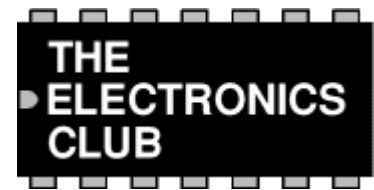


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Resistance



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Resistance

Resistance is the property of a component which **restricts the flow of electric current**. Energy is used up as the voltage across the component drives the current through it and this energy appears as heat in the component.

Resistance is measured in ohms, the symbol for ohm is an omega Ω .

1 Ω is quite small for electronics so resistances are often given in $k\Omega$ and $M\Omega$.

1 $k\Omega$ = 1000 Ω 1 $M\Omega$ = 1000000 Ω .

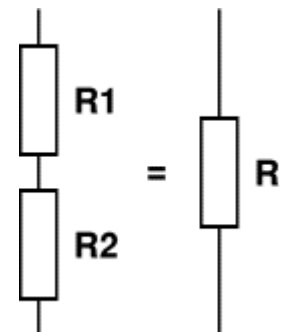
Resistors used in electronics can have resistances as low as 0.1 Ω or as high as 10 $M\Omega$.

Resistors connected in Series

When resistors are connected in series their combined resistance is equal to the individual resistances added together. For example if resistors R1 and R2 are connected in series their combined resistance, R, is given by:

Combined resistance in **series**: **$R = R1 + R2$**

This can be extended for more resistors: **$R = R1 + R2 + R3 + R4 + \dots$**

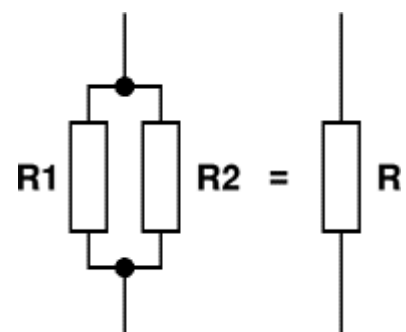


Note that the combined resistance in series will always be greater than any of the individual resistances.

Resistors connected in Parallel

When resistors are connected in parallel their combined resistance is less than any of the individual resistances. There is a special equation for the combined resistance of **two** resistors R1 and R2:

Combined resistance of **two resistors in parallel**: **$R = \frac{R1 \times R2}{R1 + R2}$**



For more than two resistors connected in parallel a more difficult equation must be used. This adds up the **reciprocal**

("one over") of each resistance to give the **reciprocal** of the combined resistance, R:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

The simpler equation for **two** resistors in parallel is much easier to use!

Note that the **combined resistance in parallel** will always be **less** than any of the individual resistances.

Conductors, Semiconductors and Insulators

The resistance of an object depends on its **shape** and the **material** from which it is made. For a given material, objects with a smaller cross-section or longer length will have a greater resistance.

Materials can be divided into three groups:

- **Conductors** which have **low resistance**.
Examples: metals (aluminium, copper, silver etc.) and carbon.
Metals are used to make connecting wires, switch contacts and lamp filaments.
Resistors are made from carbon or long coils of thin wire.
- **Semiconductors** which have **moderate resistance**.
Examples: germanium, silicon.
Semiconductors are used to make diodes, LEDs, transistors and integrated circuits (chips).
- **Insulators** which have **high resistance**.
Examples: most plastics such as polythene and PVC (polyvinyl chloride), paper, glass.
PVC is used as an outer covering for wires to prevent them making contact.

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